Abdulaziz Khader

akhader@ucsd.edu | www.linkedin.com/in/abdulaziz-khader | aokhader.vercel.app

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript / TypeScript, C#, Go

AI & ML: PyTorch, Hugging Face, scikit-learn, pandas, TensorFlow/Keras, LangChain, LangGraph, CUDA, llama.cpp, vLLM, NLTK

Web & Infrastructure: Git, React, Next.js, Node.js, FastAPI, Flask, Docker, GCP, MySQL, Supabase, SQLite, Puppeteer

EXPERIENCE

BMC Helix
Santa Clara, CA

Software Engineer — *Aug. 2026 – Present*

- Consolidated a spreadsheet-and-email mentorship process into **Skill Up**, a governed app on **BMC Helix Innovation Studio** shipped to **200+ employees**; owned **program-cycle completion, admin configuration, and notifications** across **Java/Spring, Python, Node/TypeScript** services.
- Built the **program admin dashboards** for cycle status and progress, and the **opt-out flow** letting mentors and mentees withdraw mid-cycle without stranding their counterpart.
- Cleared the pre-launch security review by hardening the deploy path: closed a **data-exposure path**, moved **hardcoded credentials into managed environment config**, and enforced **role-based access permissions** and **country-level participation rules**; added **deploy gates and test coverage** to the release pipeline, and automated **demo-data seeding** behind a deploy-script flag.

BMC Software
Santa Clara, CA

Software Engineering Intern — *June 2025 – Aug. 2025*

- Delivered a new **AIOps hub** integrating customers' **ServiceNow** data within BMC's ecosystem for **100k users** by building **LangChain-powered CRUD tools** over ServiceNow APIs, resulting in a projected **17% revenue increase.**
- Hosted the AIOps hub on an **ASGI web server** for real-time updates and optimized the current implementation to run **60% faster**.
- Engineered a **semantic model** that dynamically parses data configurations; refactored **1,000+ lines** into a reusable **tool factory** that extends to other products such as **Salesforce**.

Ocelot
Remote

AI Studio Fellow — *Aug. 2024 – Dec. 2024*

- Built a **sentiment-conditioned LLM** on student mental health conversations – **84% tone accuracy (+30% over GPT-2)**, validated through **human evaluation scoring 4.1/5.0**.
- Optimized a **7B-parameter deployment** from **14 GB to 3.5 GB** via **4-bit quantization** (98% quality preserved), enabling real-time inference on local hardware.
- Accelerated training from **8 hours to 30 minutes** through **CUDA optimization** across **5,600+ multi-domain samples**.

University of California – San Diego
La Jolla, CA

Instructional Assistant — *Sep. 2023 – June 2024*

- Mentored **800+ students** across two courses in computer systems and software tools – **C pointers, dynamic memory allocation, and bitwise operations** – tutoring one-on-one and in section.
- Authored and assembled course assignments; proctored and graded **1,000+ exams**, earning **100% positive student feedback**.

PROJECTS

GameExplorer — **Shipped to Web, iOS and Android** | *Next.js, React Native, TypeScript, Supabase* — *Mar. 2025 – Present*

- Built a **Turborepo monorepo** shipping chess, checkers, and reversi to **web, iOS, and Android** from shared TypeScript packages — **Next.js 15** and **React Native** — with **OAuth**, **Supabase** persistence, per-game rule sets, and a replay viewer.
- Ran a **20-user beta across a wide age range** that invalidated **three design assumptions** — that players knew the rules, that instant piece movement was readable, and that a dark default suited everyone — shipping a light theme, move animations with haptics, and per-game tutorials.
- Scrapped a self-built **minimax engine** after validating it at 1400+ on rated tactical positions, then watching 1400-rated players beat it consistently in full games; replaced it with **Stockfish.js** tuned by search depth and move-generation limits.

RoboRanger — **Best Edge AI & ElevenLabs Award** | *Python, C++, TFLite, llama.cpp* — *Apr. 2026 – Jun. 2026*

- Deployed a handheld, fully offline wildlife guide on an Arduino UNO Q: point it at an animal and it identifies the species and answers spoken questions aloud — classifier, speech-to-text, retrieval, a 360M LLM and text-to-speech all on-device with no network.
- Shipped the 50-species classifier after diagnosing that full-integer quantization collapsed MobileNetV3-Large from **79% to 17% top-1**; dynamic-range quantization held **79.5% top-1** at 3.4 MB, **91.1% precision**, and **0% false accepts** on empty frames.
- Cut median query-response time from **8.7s to 438ms** with a four-stage router — wildlife gate, blurb store, 8-way intent classifier, then a sub-millisecond direct route answering common intents from typed fields instead of invoking the LLM.

GPS IMU Fusion — **Merged into the official DonkeyCar** | *Python, ROS, Keras, Embedded* — *May 2026 – Jun. 2026*

- Implemented a **5-state Extended Kalman Filter** fusing BNO08X IMU with u-blox C099-F9P GPS to eliminate dead-reckoning drift on a self-driving car; refactored the sensor pipeline into vectorized tuples with **backward-compatibility layers** so legacy MPU6050/9250 sensors route into the new EKF without crashing the vehicle loop.

EDUCATION

University of California – San Diego
Graduated June 2026

Bachelor of Science in **Computer Engineering**, Minor in Mathematics — GPA: **3.7/4**